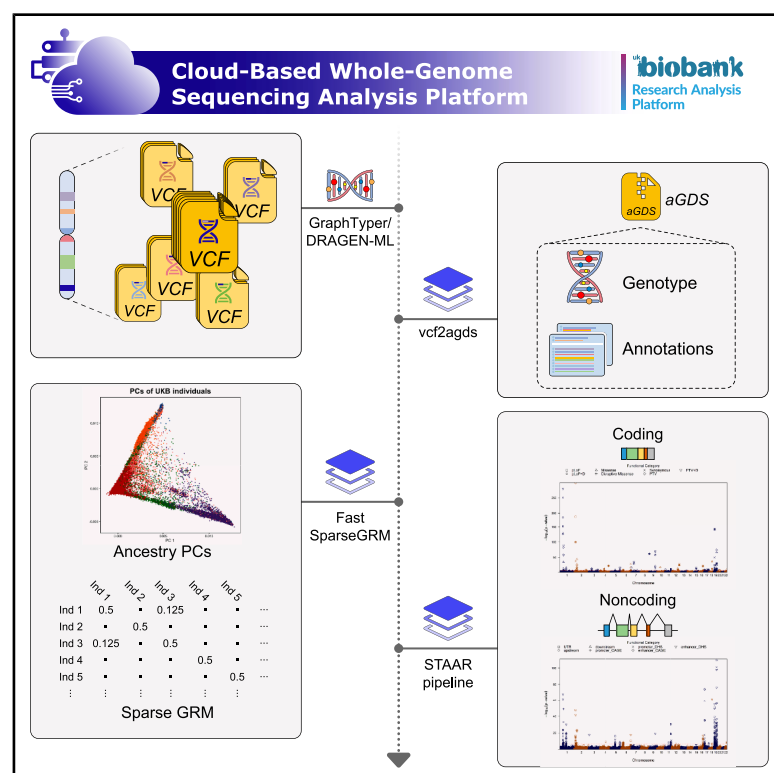


Streamlining large-scale genomic data management: Insights from the UK Biobank whole-genome sequencing data

Graphical abstract



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In brief

Li et al. present a streamlined, cloud-optimized approach for managing and analyzing large-scale whole-genome sequencing datasets. Using the UK Biobank as a case study, they introduce new tools and data formats that greatly reduce storage needs while integrating variant functional information, enabling fast, scalable, and reproducible genomic analyses and discoveries.

Highlights

- Annotated GDS (aGDS) format stores genotype and functional annotations in one file
- Efficient aGDS storage supports scalable biobank WGS data management in the cloud
- *vcf2agds* toolkit streamlines conversion from bulky VCF to analysis-ready aGDS files
- *STAARpipeline* integrates aGDS files and enables functionally informed WGS analysis



Short article

Streamlining large-scale genomic data management: Insights from the UK Biobank whole-genome sequencing data

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SUMMARY

Biobank-scale whole-genome sequencing (WGS) studies are increasingly pivotal in unraveling the genetic bases of diverse health outcomes. However, managing and analyzing these datasets' sheer volume and complexity presents significant challenges. We highlight the annotated genomic data structure (aGDS) format, substantially reducing the WGS data file size while enabling seamless integration of genomic and functional information for comprehensive WGS analyses. The aGDS format yielded 23 chromosome-specific files for the UK Biobank 500k WGS dataset, occupying only 1.10 terabytes of storage. We develop the *vcf2agds* toolkit that streamlines the conversion of WGS data from VCF to aGDS format. Additionally, the *STAARpipeline* equipped with the aGDS files enabled scalable, comprehensive, and functionally informed WGS analysis, facilitating the detection of common and rare coding and noncoding phenotype-genotype associations. Overall, the *vcf2agds* toolkit and *STAARpipeline* provide a streamlined solution that facilitates efficient data management and analysis of biobank-scale WGS data across hundreds of thousands of samples.

INTRODUCTION

Over the past decade, rapid advancements in whole-genome sequencing (WGS) technologies have generated vast amounts of data, enabling a comprehensive view of the entire genome. Notably, the National Heart, Lung, and Blood Institute (NHLBI) Trans-Omics for Precision Medicine Program (TOPMed),¹ the National Institutes of Health (NIH) All of Us Research Program,² and the UK Biobank³ have sequenced the whole genomes of approximately 218,000, 414,000, and 500,000 individuals, respectively. Together, these initiatives have identified over 1 billion genetic variants, presenting unprecedented opportunities and challenges for data management and analysis at the biobank scale. A widely used format for storing human genetic variation data is the variant call format (VCF),⁴ which contains detailed information and genotype counts for all genomic variants across study participants. For instance, the UK Biobank

released the 500k WGS jointly called variant data produced using GraphTyper (data field: 23,374) in 151,561 blocks of project VCF (pVCF) files across the genome (Table S1),⁵ with a total of 1473.85 terabyte (TiB) and an average of 9.96 gibibytes (GiB) per file. However, the large size and inherent complexity of VCF files often hinder efficient downstream analyses, particularly in biobank-scale studies such as those involving the UK Biobank.

To address the limitations of VCF files in large-scale genomic studies, several alternative data formats have been developed, including PLINK BED,⁶ BGEN,⁷ and genomic data structure (GDS)⁸ format. Among these, the SeqArray GDS⁹ format is designed as a storage-efficient, high-performance data format for WGS variant calls, utilizing a hierarchical structure to support efficient data access and seamless integration with several analytical pipelines. Building on this, the annotated GDS (aGDS) format expands the capabilities of SeqArray GDS by



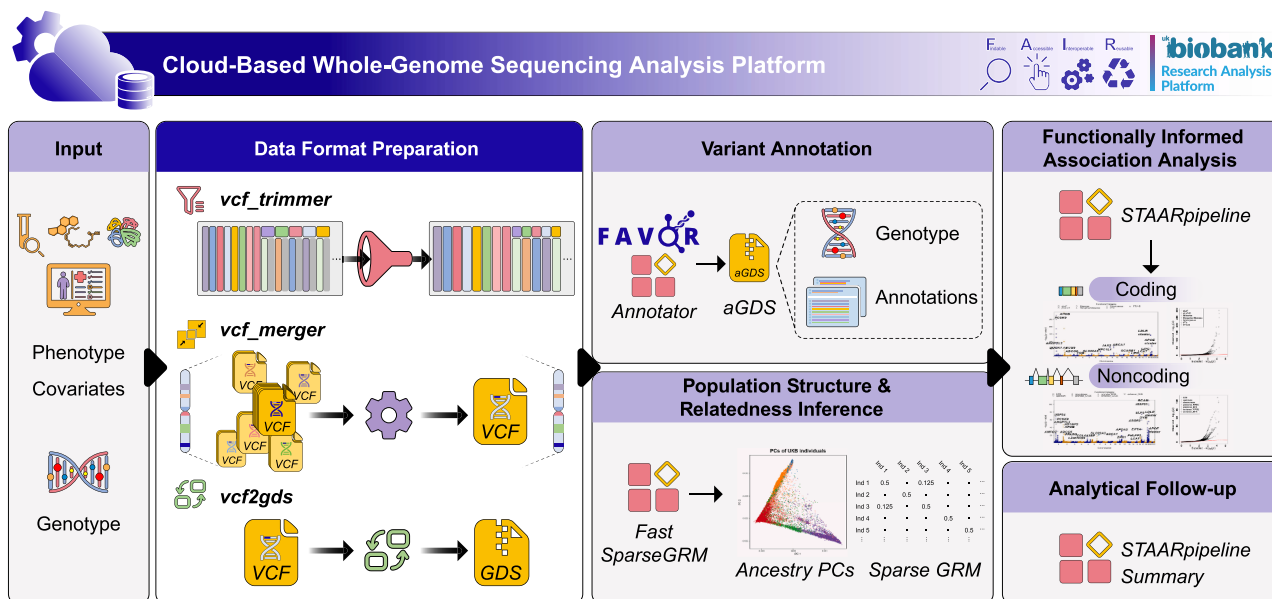


Figure 1. Overview of the cloud-based whole-genome sequencing analysis workflow

Input: prepare the input data, including genotypes, phenotypes, and covariates. Data format preparation: process whole-genome sequencing (WGS) genotype data from VCF format to GDS format using the *vcf2gds* toolkit, including *vcf_trimmer*, *vcf_merger*, and *vcf2gds* tools. Variant annotation: functionally annotate all variants in the genome and generate aGDS files using *favorannotator*. Population structure and relatedness inference: calculate ancestry principal components (PCs) and sparse genetic relatedness matrix (GRM) using *FastSparseGRM*. Functionally informed association analysis: perform functionally informed association analysis using *STAArPipeline*. Analytical follow-up: provide result summarization, visualization, and analytical follow-up using *STAArPipelineSummary*.

incorporating multi-faceted variant annotations to facilitate functionally informed downstream analysis in an all-in-one file.¹⁰

RESULTS

vcf2gds toolkit and aGDS file structure

We present the *vcf2gds* toolkit, which streamlines the conversion of WGS data from VCF to aGDS through a reproducible four-step process, including VCF trimming, VCF merging, format conversion, and variant annotation (Figure 1). The first step (*vcf_trimmer*) involves reducing the complexity of VCF files using *bcftools*¹¹ by applying quality filters, such as excluding low-quality variants with an AAScore of less than 0.5 assigned by GraphTyper,^{5,12} and removing non-essential fields, such as the Phred-scaled genotype likelihoods (FORMAT/PL) and read depth of multiple alleles (FORMAT/MD), that will not be used by downstream analyses. This preprocessing step minimizes file sizes and improves compatibility for merging, and users can customize this step by selecting specific fields and thresholds to retain the most relevant information for analysis. After trimming, the second step (*vcf_merger*) is to merge multiple VCF files into a single dataset for each chromosome. Efficient merging ensures that the subsequent conversion to other file formats is streamlined and manageable, which is particularly critical for large-scale projects like the UK Biobank. The third step (*vcf2gds*) converts the merged VCF files into GDS files using SeqArray,⁹ which allows for efficient access to specific sample and/or variant subsets without loading the entire dataset into memory. The last step (*favorannotator*) performs functional annotation of the SeqArray GDS files into aGDS files through

the Functional Annotation of Variants Online Resource (FAVOR) database.¹³

The resulting aGDS files are equipped with both genotype data and variant annotations in a unified hierarchical structure. Specifically, genotype information is stored using a highly compact 2-bit array format, organized across the dimensions of ploidy, sample, and variant. Each genotype array utilizes the Lempel-Ziv Markov chain (LZMA) lossless compression algorithm, which performs LZMA compression coupled with independent indexing to facilitate efficient random access.⁹ An accompanying index vector is employed to precisely indicate the number of bits assigned per genotype entry, enabling rapid retrieval and decompression. Alongside genotype data, aGDS files simultaneously embed multi-dimensional functional annotations, including predicted variant functional impacts, regulatory element associations, protein functional annotations, and evolutionary conservation scores, stored using the LZMA lossless compression algorithm. Detailed annotations could be found in Table S2. The integrated and hierarchically structured storage scheme of aGDS format optimizes computational efficiency, enabling the simultaneous querying and retrieval of genotype and matched annotation data, thereby significantly reducing computational overhead for functionally informed downstream genomic analyses.

Benchmarking and application to UK Biobank WGS data

In a case study using the UK Biobank WGS data, the original pVCF files (data field: 23,374) of 500k individuals, provided by the UK Biobank in 151,561 blocks, totaled 1,473.85 TiB. After applying the trimming step of the *vcf2gds* toolkit, the VCF file

Table 1. Summary of the UK Biobank 500k WGS pVCF files and generated aGDS files

Chr	VCF files		aGDS files		Compression rate
	Total count	Total size (TiB)	Total count	Total size (TiB)	
1	24,896	114.08	1	0.08	1,349.42
2	24,220	124.06	1	0.09	1,398.73
3	19,830	102.28	1	0.07	1,376.76
4	19,022	98.74	1	0.07	1,338.72
5	18,154	92.21	1	0.07	1,364.03
6	17,082	86.54	1	0.07	1,268.16
7	15,936	82.06	1	0.06	1,322.19
8	14,514	78.28	1	0.06	1,368.84
9	13,840	63.62	1	0.05	1,343.97
10	13,380	68.44	1	0.05	1,298.22
11	13,510	69.22	1	0.05	1,328.04
12	13,328	67.15	1	0.05	1,320.88
13	11,438	49.52	1	0.04	1,317.16
14	10,706	45.81	1	0.03	1,327.24
15	10,200	42.46	1	0.03	1,329.15
16	9,034	46.05	1	0.03	1,350.84
17	8,326	40.79	1	0.03	1,278.62
18	8,038	38.95	1	0.03	1,334.02
19	5,862	30.71	1	0.03	1,169.46
20	6,446	32.20	1	0.02	1,310.84
21	4,672	19.28	1	0.02	1,219.98
22	5,082	19.91	1	0.02	1,193.58
X	15,606	61.52	1	0.04	1,501.72
Total	303,122	1,473.85	23	1.10	1,336.35

VCF files: information for all pVCF and index files, including the total number of files and their combined size in terabytes (TiB). aGDS files: information for generated aGDS files, including the total number of files and their combined size in terabytes (TiB). Compression rate: file size ratio of VCF and aGDS file.

sizes decreased dramatically to 5.99 TiB, representing a substantial reduction in data size. Following format conversion using SeqArray, the resulting GDS files were further compressed to only 0.89 TiB (Table S1). Finally, after the addition of detailed functional annotations, the converted aGDS files were stored as one file per chromosome, achieving a total size of only 1.10 TiB, representing a 1,336 $\times$ reduction compared to the original pVCF files (Table 1). We further benchmarked the storage requirements of aGDS files using the 200k release of the UK Biobank WGS dataset, compared with pVCF files (data field: 24,304), PLINK BED files (data field: 24,305), and BGEN files (data field: 24,306). The original pVCF files, PLINK BED files, and BGEN files totaled 533.62, 26.86, and 0.87 TiB in size, respectively. The aGDS files, generated using *vcf2agds*, had a total size of 0.49 TiB, representing a 1,084 $\times$ reduction in size compared to the original pVCF files and a compression ratio of 54.61 $\times$ relative to PLINK BED files and 1.77 $\times$ relative to BGEN files. Notably, the aGDS format uniquely integrates variant functional annotation data in the same file, making it the most

compact and information-rich data structure among the formats compared (Table S3). These results underscore the flexibility and efficiency of the aGDS format in managing large-scale genomic datasets, which is only required to be performed once for all downstream analyses.

The aGDS format supports a variety of genomic analysis tools, such as *FastSparseGRM*¹⁴ for efficient calculation of ancestry principal components (PCs) and sparse genetic relatedness matrix (GRM) to account for population structure and sample relatedness; *GENESIS*¹⁵ and *SAIGEgds*¹⁶ for genome-wide association studies; and *RICE*¹⁷ for robust polygenic risk score construction. Biobank-scale WGS studies present a unique opportunity to investigate the contribution of coding and noncoding rare variants to complex human diseases and traits. *STAARpipeline*¹⁸ efficiently performs functionally informed WGS association analyses by leveraging the integrated genotype and functional annotation data uniquely provided by aGDS files, with enhanced interpretability and significantly reduced computational overhead. Specifically, *STAARpipeline* encompasses both single-variant and variant-set approaches across a variety of coding and noncoding genomic units (Figure 1).

For single-variant analysis, *STAARpipeline* provides valid individual *p* value for each genetic variant. For variant-set analysis, *STAARpipeline* provides seven coding and seven noncoding functional categories (masks) for each protein-coding gene and one functional category for noncoding RNA (ncRNA) genes.¹⁸ The coding masks include putative loss-of-function (pLoF; stop gain, stop loss, and splice), protein-truncating (PTV; stop gain, stop loss, splice, frameshift deletion, and frameshift insertion), missense, disruptive missense, combinations of pLoF/PTV and disruptive missense, and synonymous variants. The variant classes are defined according to GENCODE Variant Effect Predictor (VEP) annotations, with disruptive missense variants further defined by MetaSVM, indicating their potential deleteriousness. Noncoding masks comprise variants located in promoter or enhancer regions overlapping cap analysis of gene expression (CAGE) or DNase I hypersensitivity (DHS) sites, in untranslated regions (UTRs), and upstream or downstream regions of protein-coding genes, as well as ncRNA genes. Promoter variants are defined within ± 3 kilobase (kb) windows of transcription start sites overlapping with CAGE or DHS sites, while enhancer variants are defined using GeneHancer-predicted regions overlapping with CAGE or DHS sites. We define the UTR, upstream, downstream, and ncRNA variants using GENCODE VEP categories, specifically including variants from both 5' and 3' UTRs and exonic and splicing regions of ncRNA genes, respectively (Table S2).

Furthermore, *STAARpipeline* enhances analysis power by incorporating multi-faceted quantitative functional annotations, such as the annotation PCs (aPCs) (Table S2), through a dynamic weighting scheme using the STAAR framework.¹⁰ These analyses further take advantage of the capability of aGDS files, which enable efficient access to sparse genotype dosages for any set of samples and/or variants and simultaneous retrieval of corresponding variant annotations.^{9,13} The follow-up tool, *STAARpipelineSummary*, provides comprehensive result summarization, visualization, and conditional analysis (Figure 1),

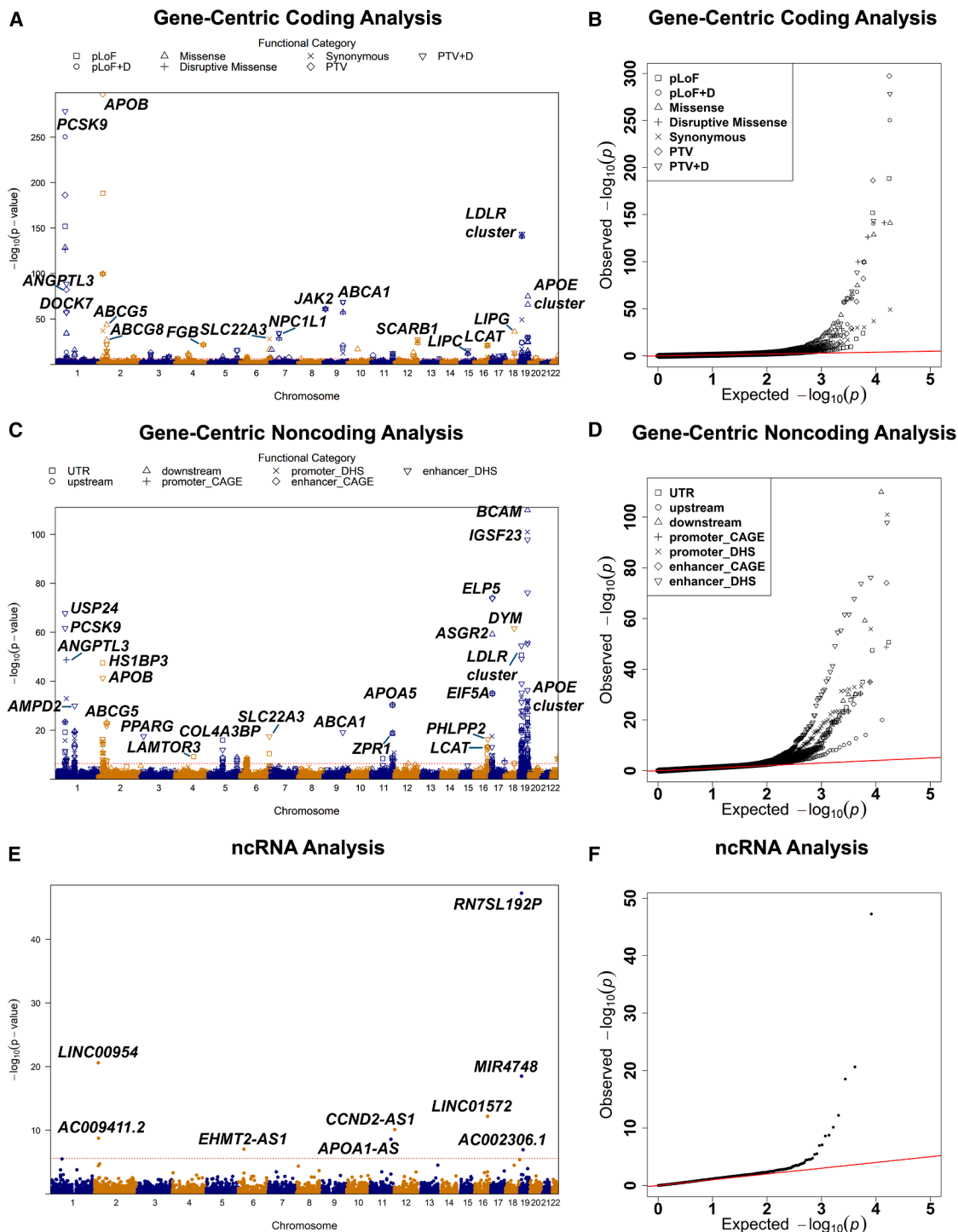


Figure 2. Manhattan plots and Q-Q plots for gene-centric coding analysis and noncoding analysis of total cholesterol

(A) Manhattan plots for gene-centric coding analysis. The horizontal line indicates a genome-wide STAAR-O p value threshold of 3.57×10^{-7} . The significant threshold is defined by multiple comparisons using the Bonferroni correction ($0.05/(20,000 \times 7) = 3.57 \times 10^{-7}$).

(B) Quantile-quantile (Q-Q) plots for gene-centric coding analysis.

(C) Manhattan plots for gene-centric noncoding analysis. The horizontal line indicates a genome-wide STAAR-O p value threshold of 3.57×10^{-7} . The significant threshold is defined by multiple comparisons using the Bonferroni correction ($0.05/(20,000 \times 7) = 3.57 \times 10^{-7}$).

(D) Q-Q plots for gene-centric noncoding analysis.

(legend continued on next page)

enabling the detection of putatively novel associations by adjusting for previously reported variants and accounting for independent associations via aggregate conditioning.^{18,19} Collectively, these features underscore *STAARpipeline*'s ability to fully leverage the extensive annotations and computational efficiency provided by aGDS files.

As a demonstration of the utility of aGDS files, we conducted a WGS association analysis of total cholesterol (TC) using variants that passed quality control in 500k WGS data (Table S4). Leveraging the *STAARpipeline* for variant-set analysis across diverse coding and noncoding genomic units¹⁸ (STAR Methods), we identified 480 genome-wide significant associations. These associations were determined using Bonferroni-corrected significance thresholds of $\alpha = 0.05/(20,000 \times 7) = 3.57 \times 10^{-7}$, accounting for 7 different coding or noncoding masks across protein-coding genes, and $\alpha = 0.05/20,000 = 2.50 \times 10^{-6}$, accounting for ncRNA genes. Among these, 200 associations were uncovered through gene-centric coding analyses, while 280 were identified through gene-centric noncoding analyses (Figure 2; Tables S5 and S6).

DISCUSSION

The aGDS format provides a streamlined solution for efficient management and analysis of biobank-scale WGS data of hundreds of thousands of samples. As WGS datasets continue to expand in size and complexity, our proposed tools will be increasingly useful for unlocking the full potential of genomic research. Centralized efforts are encouraged to harmonize the WGS data for large-scale consortia and biobanks and make aGDS files available to all potential users. The aGDS files have already been generated and widely used by the NHLBI TOPMed Program and are accessible through the NHLBI BioData Catalyst platform.

Limitations of the study

Despite their advantages, the aGDS format and associated genomic analysis toolkit have certain limitations. When applying the aGDS format and *STAARpipeline* to other large-scale resources, such as the NIH All of Us Research Program or other international biobanks, additional steps may be needed. For example, the All of Us short-read WGS data are stored in the Hail VariantDataset (VDS) format. One needs to convert VDS files to VCF files before applying the *vcf2agds* toolkit. Careful quality control, standardization, and validation steps are thus recommended to ensure consistent results across multiple datasets with differing sequencing protocols, variant calling pipelines, or annotation standards.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Zilin Li (lizi@nenu.edu.cn).

Materials availability

No materials were generated in this study.

Data and code availability

vcf2agds is a suite of open-source software freely available at https://github.com/drawood/vcf2agds_overview. Access to the UK Biobank resource is available via application (<https://www.ukbiobank.ac.uk/>). The development of software was undertaken under UK Biobank applications 9072 and 103356. Benchmarking results for UK Biobank WGS data were generated using the UK Biobank resource under applications 91486, 52008, and 211447. All analyses were conducted in the UK Biobank Research Analysis Platform (RAP; <https://ukbiobank.dnanexus.com/>), and the generated aGDS files will be made available as a "returned dataset." *FastSparseGRM* is implemented as an open-source R package available at <https://github.com/rounakdey/FastSparseGRM>. *STAAR* is implemented as an open-source R package available at <https://github.com/xihaoli/STAAR>. *STAARpipeline* and *STAARpipelineSummary* are implemented as open-source R packages available at <https://github.com/xihaoli/STAARpipeline> and <https://github.com/xihaoli/STAARpipelineSummary>. *STAARpipeline* is implemented as an applet in the UK Biobank RAP available at <https://github.com/xihaoli/staarpipeline-rap>.

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AUTHOR CONTRIBUTIONS

Xihao Li, A.R.W., X. Lin, and Z.L. designed the experiments. Xihao Li, A.R.W., Y.Y., M.Z., Y.H., and Z.L. performed the experiments. Xihao Li, A.R.W., Y.Y., M.Z., Y.H., G.H., R.N.B., M.N.W., W.L., Xiaoyu Li, X. Lin, and Z.L. acquired, analyzed, or interpreted data. Xihao Li, A.R.W., and Z.L. developed and maintained the software. Xihao Li, A.R.W., Y.Y., X. Lin, and Z.L. drafted the manuscript and revised it according to co-authors' suggestions. All authors critically reviewed the manuscript, suggested revisions as needed, and approved the final version.

DECLARATION OF INTERESTS

X. Lin is a consultant of AbbVie Pharmaceuticals and Verily Life Sciences.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- METHOD DETAILS
 - Analysis of total cholesterol (TC) phenotype in the UK Biobank 500k WGS data
 - Genome build

(E) Manhattan plots for ncRNA analysis. The horizontal line indicates a genome-wide STAAR-O p value threshold of 2.50×10^{-6} . The significant threshold is defined by multiple comparisons using the Bonferroni correction ($0.05/20,000 = 2.50 \times 10^{-6}$).

(F) Q-Q plots for ncRNA analysis.

In (A), (C), and (E), the chromosome numbers are indicated by the colors of dots. In (A)–(F), STAAR-O is a two-sided test.

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
UK Biobank phenotypic data	UK Biobank	Application: 52008, 91486, 211447
UK Biobank whole genome sequence data (500,000 release produced using GraphTyper)	UK Biobank	Data Field: 23374
UK Biobank whole genome sequence data (200,000 release in pVCF format)	UK Biobank	Data Field: 24304
UK Biobank whole genome sequence data (200,000 release in PLINK BED format)	UK Biobank	Data Field: 24305
UK Biobank whole genome sequence data (200,000 release in BGEN format)	UK Biobank	Data Field: 24306
Functional annotation data	FAVOR database (https://favor.genohub.org/)	https://doi.org/10.7910/DVN/1VGTJI
Software and algorithms		
vcf2agds	https://github.com/drawood/vcf2agds_overview	v1.0.0
FastSparseGRM	https://github.com/rounakdey/FastSparseGRM	Retrieved on June 2024
STAAR	https://github.com/xihaoli/STAAR	v0.9.7
STAARpipeline	https://github.com/xihaoli/STAARpipeline	v0.9.7
STAARpipelineSummary	https://github.com/xihaoli/STAARpipelineSummary	v0.9.7
STAARpipeline UK Biobank Research Analysis Platform Applet	https://github.com/xihaoli/staarpipeline-rap	v0.9.7

METHOD DETAILS

Analysis of total cholesterol (TC) phenotype in the UK Biobank 500k WGS data

We used the generated aGDS files for WGS data of 490,549 UK Biobank participants from the pVCF files (Data Field: 23374) and followed the same quality control procedure in the previous study of UK Biobank WGS data.¹² We kept all variants indicated by the FILTER label and AAScore greater than 0.5, where AAScore was generated by GraphTyper,⁵ the software used by the UK Biobank to perform genotype calling. Variants were further filtered by excluding those that failed the Hardy-Weinberg equilibrium test ($p < 1 \times 10^{-100}$) or had a call rate below 90%.^{3,20} We harmonized the total cholesterol trait (Data Field: 30690), where TC was adjusted by dividing the value by 0.8 among individuals reporting lipid lowering medication use or statin use (Data Field: 20003). A total of 462,458 individuals had data on TC.

We fit a linear regression model adjusting for age, age², sex, and the first ten ancestral principal components. Residuals were then rank-based inverse-normal transformed and multiplied by the standard deviation. We next fit a linear mixed model (LMM) for the rank normalized residuals of TC, adjusting for age, age², sex, and the ten ancestral principal components, and a variance component for an empirically-derived sparse genetic relatedness matrix (GRM) using *FastSparseGRM*¹⁴ to account for population structure and sample relatedness.

We next applied *STAARpipeline*¹⁸ to perform variant set analyses for rare variants (MAF <1%), including gene-centric analysis of protein-coding genes using seven coding variant functional categories; seven noncoding variant functional categories and rare variants in noncoding RNA (ncRNA) genes. Our analysis was performed on the UK Biobank Research Analysis Platform (RAP).

In the gene-centric coding analysis, 200 genome-wide significant associations were identified at the Bonferroni-corrected level $0.05/(20,000 \times 7) = 3.57 \times 10^{-7}$ (Figures 2A and 2B; Table S4). These associations included well-known genes such as *PCSK9*, *APOB*, the *LDLR* cluster, and the *APOE* cluster. For the gene-centric noncoding analysis of protein-coding genes, 271 genome-wide significant associations were detected at the Bonferroni-corrected level $0.05/(20,000 \times 7) = 3.57 \times 10^{-7}$ (Figures 2C and 2D; Table S5). Among these, notable associations involved genes such as *USP24*, *PCSK9*, *APOB*, the *LDLR* cluster, and the *APOE* cluster. In the gene-centric noncoding analysis of ncRNA genes, we identified 9 significant associations at the genome-wide significance level $0.05/20,000 = 2.50 \times 10^{-6}$ (Figures 2E and 2F; Table S5).

Genome build

All genome coordinates are given in NCBI GRCh38/UCSC hg38.